

Acid base imbalance

- acidemia } current states in which
- alkalemia } the blood pH is abn^o
high or low

- acidosis } the process in which
- alkalosis } the H⁺ ion concⁿ
↑ses or decreases

* Henderson-Hasselbach eqⁿ:-

$$pH = pK + \log_{10} \frac{[HCO_3^-]}{[pCO_2 \times 0.03]}$$

* Step I:-

- History & examination
- most important step
- frequently overlooked
- correlate with other findings

* Step II:-

Determine the primary disorder

- n^o values:-

n^o pH : 7.40 ± 0.02

p_aCO₂ : 38 ± 2 mmHg

[HCO₃⁻] = 24 ± 2 mmHg

- pH compatible 2 life: 6.8 to 7.8

- If pH > 7.4 ⇒ 1^o alkalemia

- If pH < 7.4 ⇒ 1^o acidemia

→ If pH & p_aCO₂ move in same direction → 1^o respi

→ If pH & p_aCO₂ move in opp. direction → 1^o metabolic

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Acid base imbalance

* Metabolic acidosis:-

$$pH < 7.38$$

$$[HCO_3^-] < 22 \text{ mmol/l}$$



Expected 2° resp compensation:

Winter's formula:-

$$PaO_2 = [HCO_3^-] + 15$$

- Complete secondary response takes 12-24 hours

* Metabolic alkalosis:-

$$pH > 7.42$$

$$[HCO_3^-] > 26 \text{ mmol/l}$$



Expected 2° resp response:

$$PaO_2 = [HCO_3^-] + 15$$

- Complete secondary response takes in 24-36 hours

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Acid base imbalance

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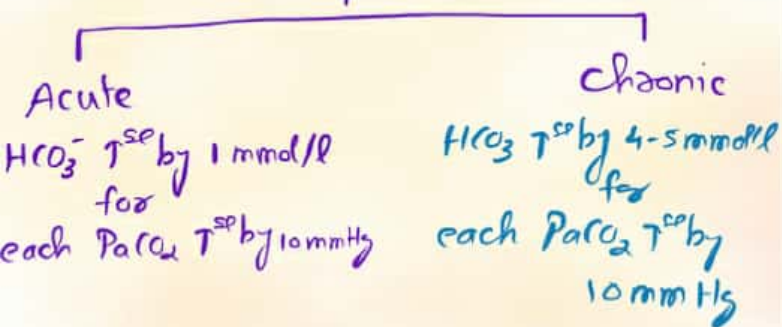
* Respiratory acidosis

- pH < 7.38

$\nwarrow$
 $P_{aCO_2} > 42 \text{ mmHg}$



2° metabolic response



- Complete response in
2-5 days

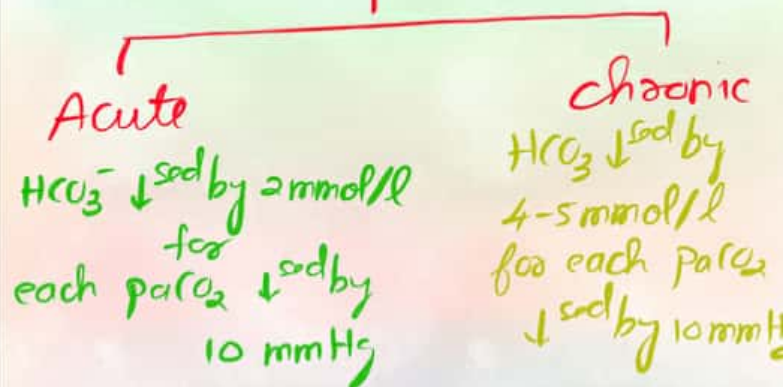
* Respiratory alkalosis:-

pH > 7.42

$\nwarrow$
 $P_{aCO_2} < 38 \text{ mmHg}$



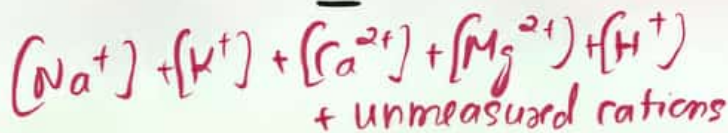
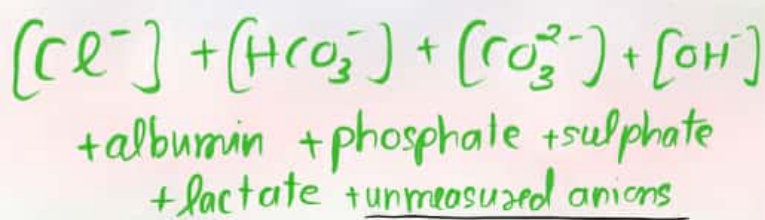
2° metabolic response



- Complete 2° response in
2-5 days

Acid base imbalance

* Step III :- Evaluation of the metabolic component



∴ unmeasured anions

$$= [Na^+] - [Cl^-] - [HCO_3^-]$$
 anion gap
 [rest negligible, can be ignored]

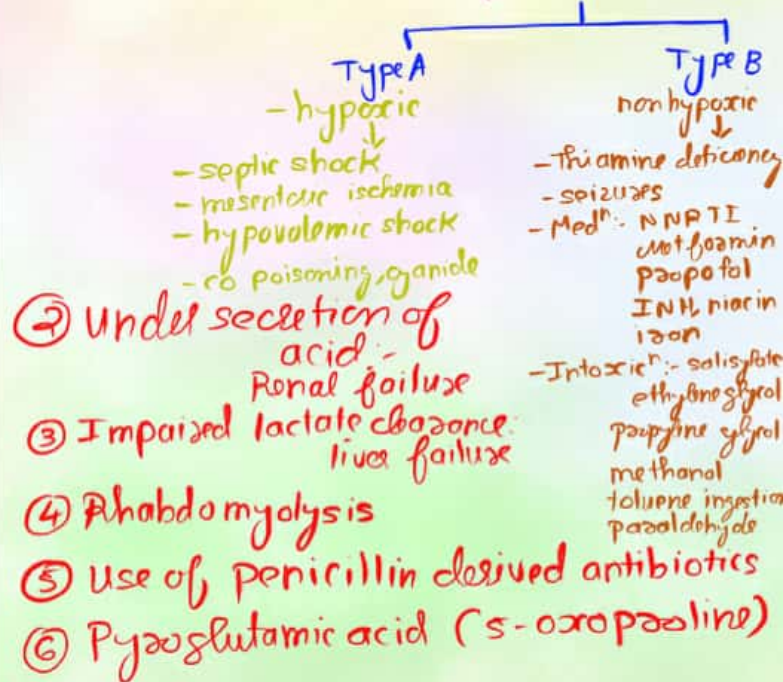
② :- $10 \pm 3 \text{ mEq/l}$

→ albumin correction:
for fall of 1g/dl of albumin
below 4g/dl, $\downarrow$ in AG by 2.5 mEq/L
 $\therefore$ corrected A.G. = A.G. + 2.5
(4 - albumin)

* Causes of HAGMA

① Over prodⁿ of acid

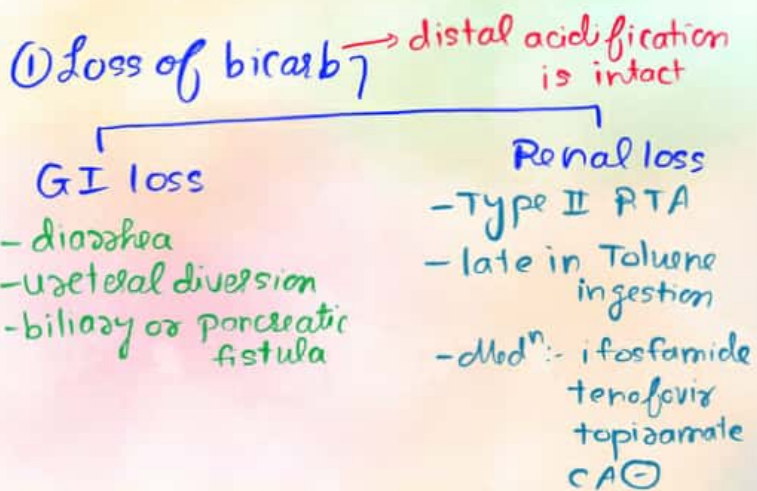
- ketoacidosis
- lactic acidosis
 - d-lactic acidosis
 - l-lactic acidosis



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Acid base imbalance

* Causes of NAGMA:-

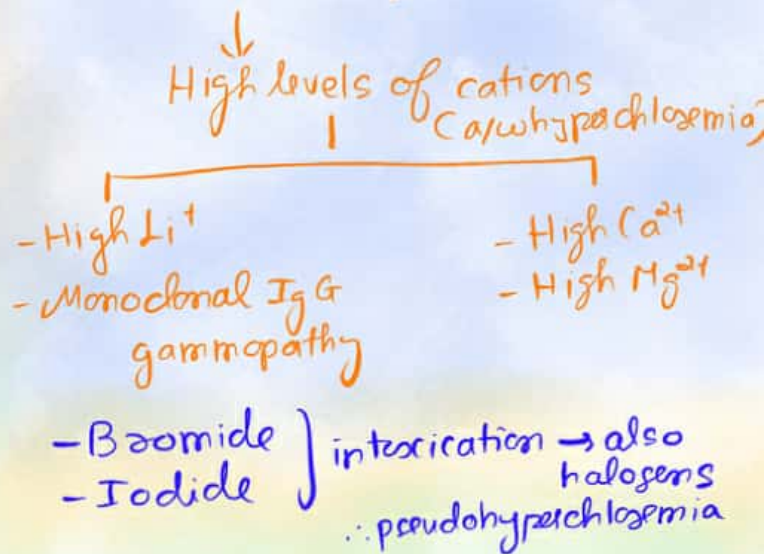


② $\downarrow$ ^{sed} Renal acid excretion:-

- | | |
|--------------------------|------------------------------------|
| ① early uraemic acidosis | } distal acidification is affected |
| ② Type I RTA | |
| ③ Type II RTA | |

$\rightarrow$ $\uparrow$ ^{se} in anion gap
 $=$ $\uparrow$ ^{sed} in unmeasured anions like lactate & acetate

* -ve anion gap



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Acid base imbalance

* Urinary A.G.:-

- hyperchloraemic acidosis

↓
↑^{sed} NH_4^+ ion excretion

↓
Can be used to differentiate
both renal & extra renal causes
of NAGMA

- U.AG is a surrogate marker of
 U.NH_4^+

- NAGMA — [loss of bicarb: NH_4^+ excretion ⊕
↓^{sed} renal acid excretion → ↓^{sed} NH_4^+ excretion

$$\text{U.AG} = (\text{U.Na}^+ + \text{U.K}^+) - (\text{U.Cl}^-)$$

+ve

⊕

-ve → distal acidⁿ intact

- Renal failure
- Type I RTA
- Type II RTA

- diarrhoea
- Type II RTA
- ⊕ healthy

↓
↑^{sed} NH_4^+ excretion

Concept:-

NH_4^+ excretion is always
followed by Cl^- excretion

↓

∴ ↑ NH_4^+ excretion

↓

↑^{sed} Cl^- excretion

↓

↑^{sed} U. Cl^-

↓

-ve U. Anion gap

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Acid base imbalance

* Step IV:-

- Calculate delta-delta (Δ-Δ) only if HAGMA

$$-\Delta-\Delta = \text{Na}^+ - (\text{Cl}^- + 36)$$

(Dalal's formula)

> 6



superadded
metabolic
alkalosis

< 6



superadded
metabolic
acidosis

* Step V:-

- Calculate S. osmolal gap

- Difference betⁿ
measured S. osmolality
←
calculated S. osmolality

$$(2 \times \text{S. Na}^+) + \frac{\text{S. glucose}}{18} + \frac{\text{BUN}}{2.8}$$

- If difference > 10 mOsm/Kg



Toxic osmolal
alcohol ingestion

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Acid base imbalance

* Step VI :-

- alveolar-arterial O_2 tension difference
(A-a)

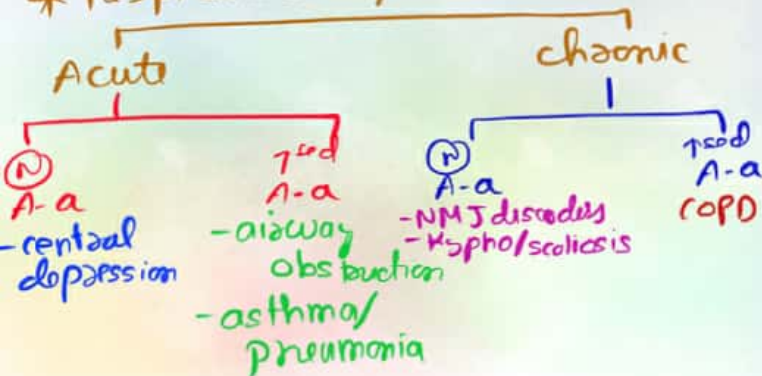
- (N) : for every decade a person has lived, (A-a) difference

$\uparrow^{ses}$ by 2 mmHg

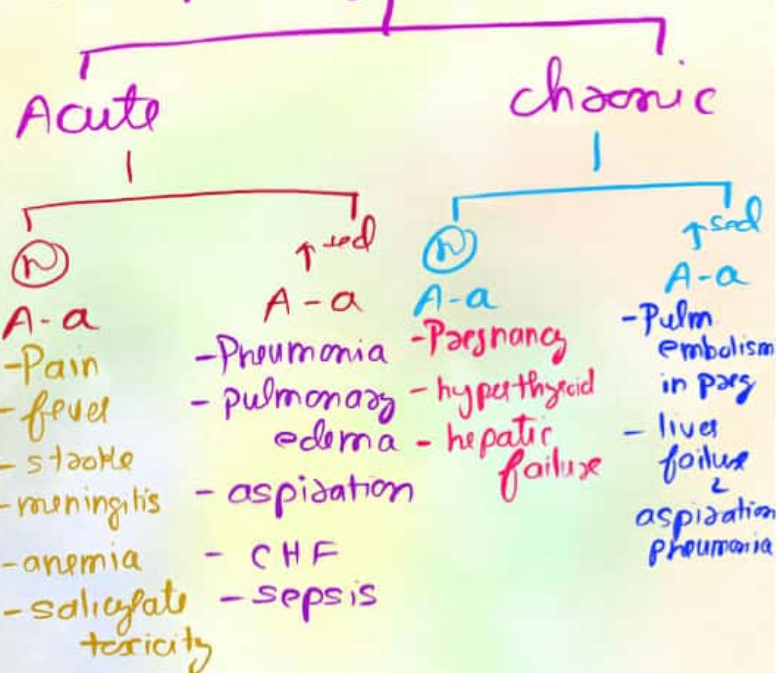
- expected A-a difference = $\frac{Age}{4} + 4$

- If A-a $\uparrow^{sed}$ $\rightarrow$ 1^o lung condition

* Respi acidosis



* Respiratory alkalosis



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